

UNIVERSITY OF OSLO

A Self-Adaptive Digital Twin Architecture for Dynamic Resource Management

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SEAMS 2026

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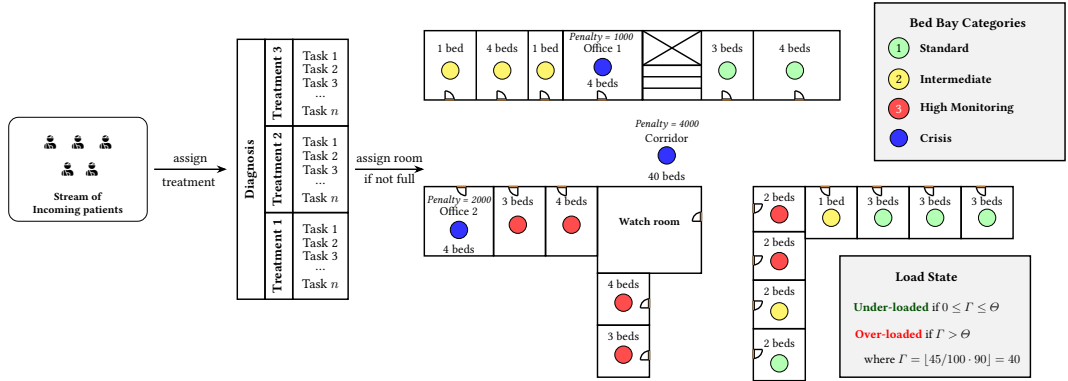
April 13, 2026



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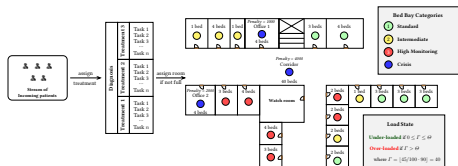
- ① Rationale
- ② Digital Twin Concepts
- ③ Digital Twin Architecture

Why do we need it?



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- Hospital demand changes quickly; resource capacity does not

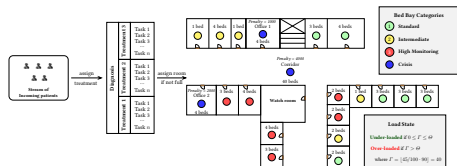


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Why do we need it?

- Hospital demand changes quickly; resource capacity does not
- Admission is often handled manually
- Depends on several factors
- Static architectures struggle when the underlying structure changes



Baseline: BedreFlyt and the gap

- **BedreFlyt** is a hospital-ward digital twin that combines an evolving semantic model, simulation-driven forecasting, and optimisation to recommend bed-bay allocations to staff

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- It produces bed-bay recommendations for human operators
- Human-in-the-loop actions can diverge from DT suggestions
- **Main gap:** the original ward layout is treated as fixed

What is the goal

Goal: enable the digital twin to reconfigure ward capacity at runtime (open/close spare rooms) in response to changing patient demand, while staying aligned with the real ward and keeping a human in the loop.

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- Human-in-the-loop supervision over the adaptation loop

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- 2 Digital Twin Concepts
 - Modelling aspects
- 3 Digital Twin Architecture

What is a Digital Twin

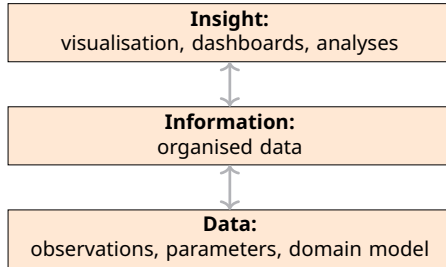
Digital twins are, as defined by NASEM

NASEM Definition of DT (2024)

A digital twin is a set of virtual information constructs that mimics the structure, context, and behaviour of a natural, engineered, or **social system** (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and **informs decisions** that realise value. The bidirectional interaction between the virtual and the physical is central to the digital twin^a.

^aNational Academies of Sciences, Engineering, and Medicine (NASEM) (2024): Foundational Research Gaps and Future Directions for Digital Twins. The National Academies Press

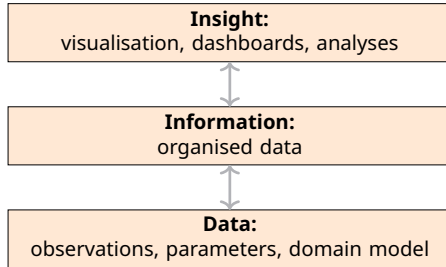
Digital Twins components



Capabilities for different insights:

- **Descriptive:** Insight into the past ("what happened")

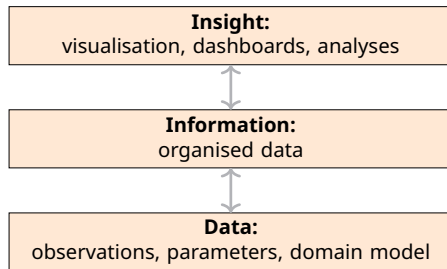
Digital Twins components



Capabilities for different insights:

- **Descriptive:** Insight into the past (“what happened”)
- **Predictive:** Understand the future (“what may happen”)

Digital Twins components



Capabilities for different insights:

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- **Prescriptive:** Advise on possible outcomes ("what if / what should")

Digital Twins components

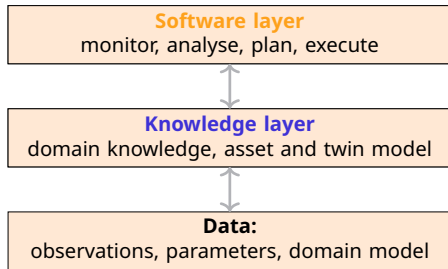
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What next:

- **Reactive:** Automated decision making

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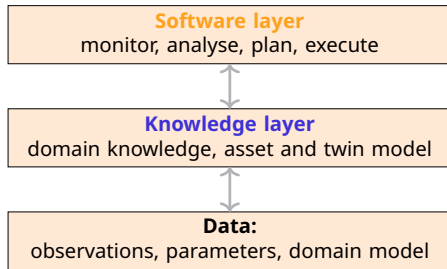
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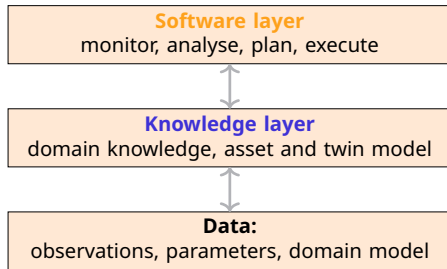
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- DT must adapt both structurally and behaviourally
- States of the DT cycles over time and they need to be captured

Twinning in SMOL

Programming support for twins with a behavioural and a structural layer
smolang.org



Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

SMOL: Semantic Model Object Language

- Smalll OO programming system
- Ontology reasoners allow querying the KB
- Used to create the digital model

smolang.org



Twinning in SMOL

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behavioural twins in SMOL

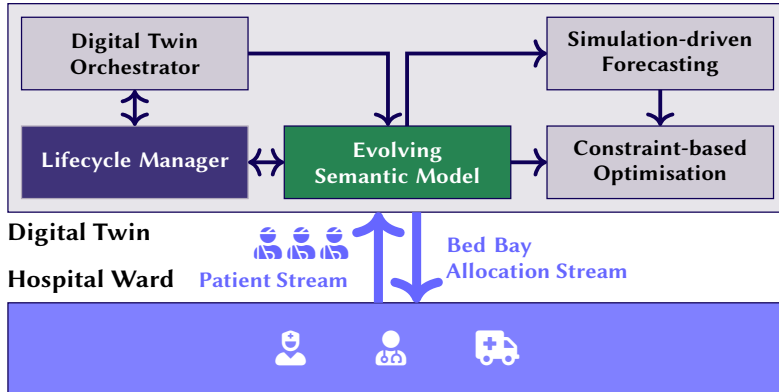
- SMOL can encapsulate (simulation) models based on the FMI standard
- Can automatically adapt the object states in the KB at runtime through semantic lifting



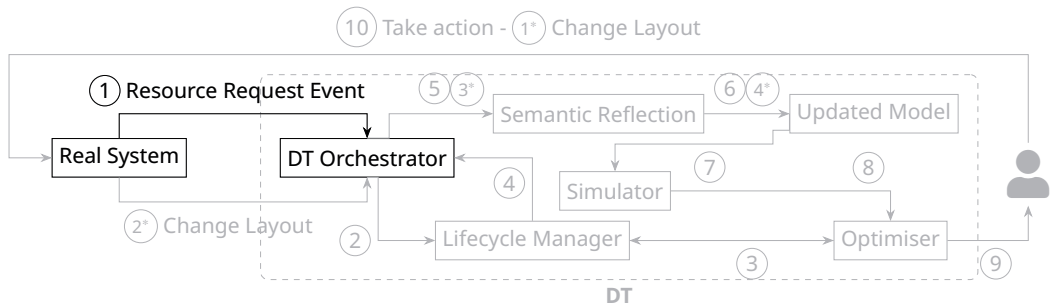
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 - Bedreflyt DT Execution

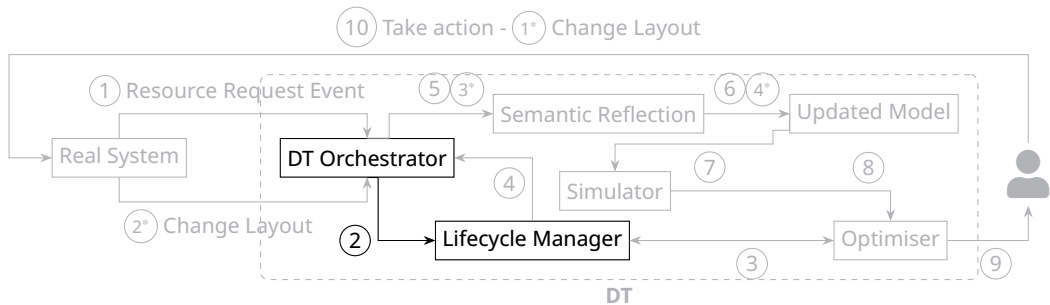
Solution: DYNRESDT (Self-Adaptive DT Architecture)



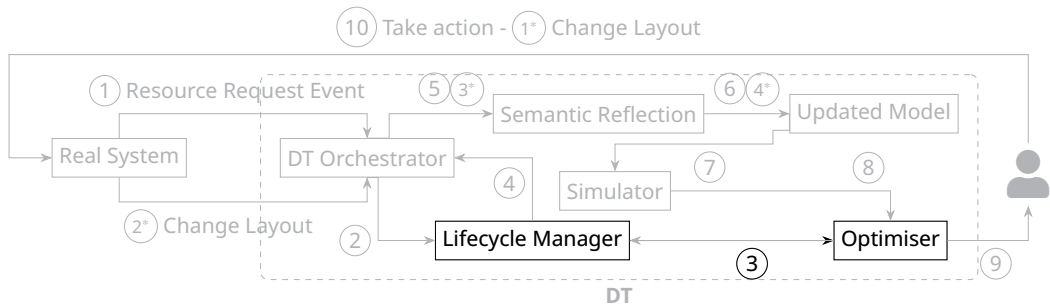
Human-in-the-loop



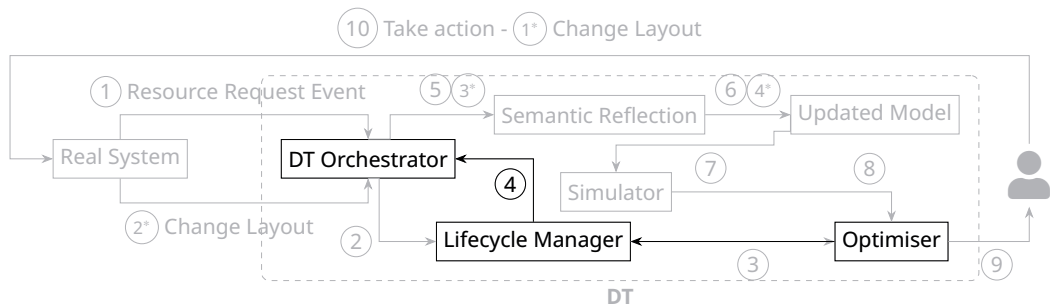
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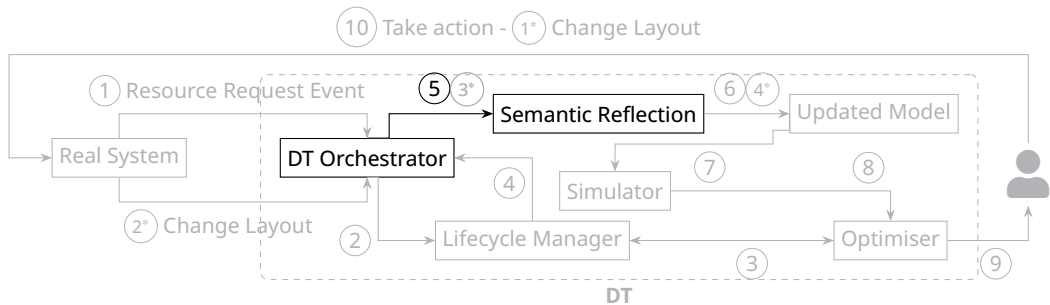
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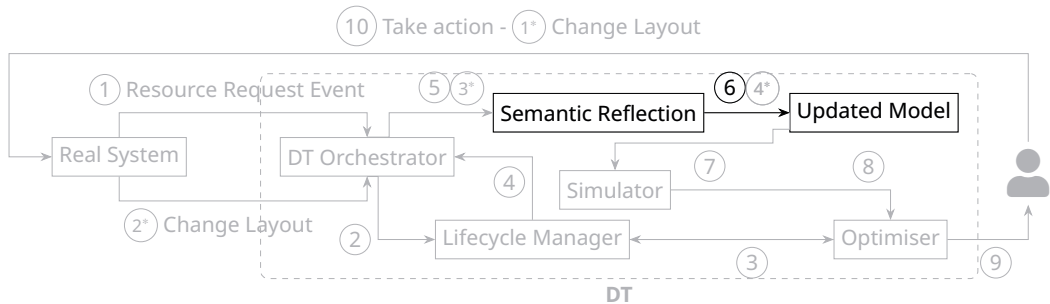
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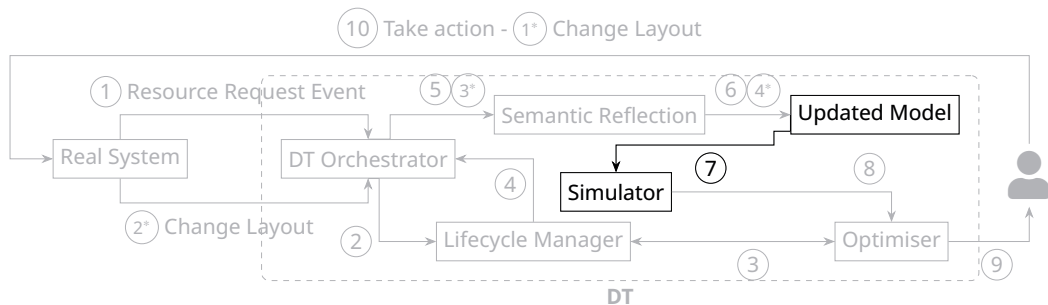
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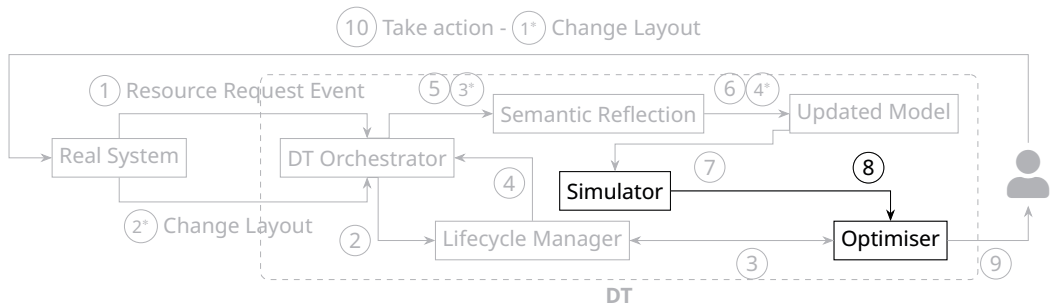
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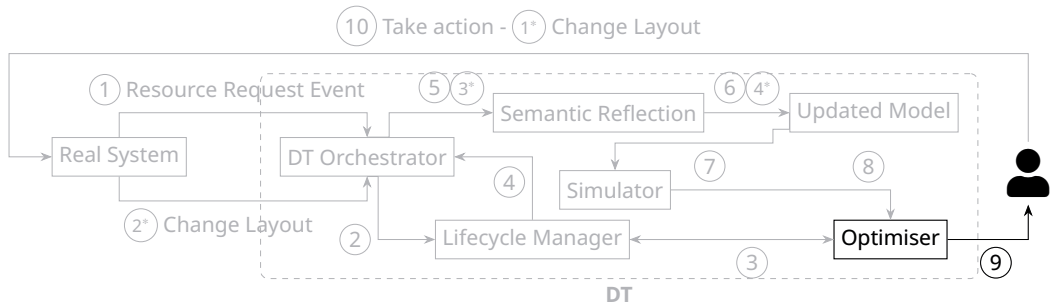
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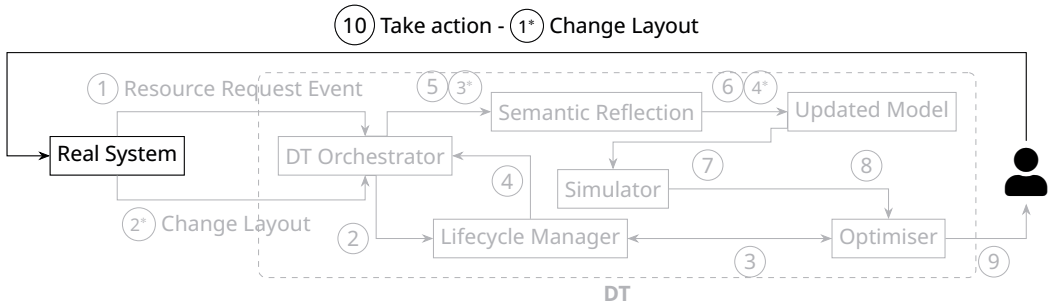
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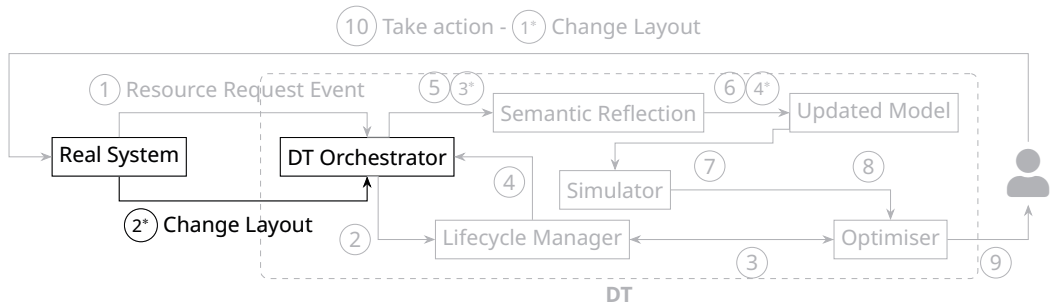
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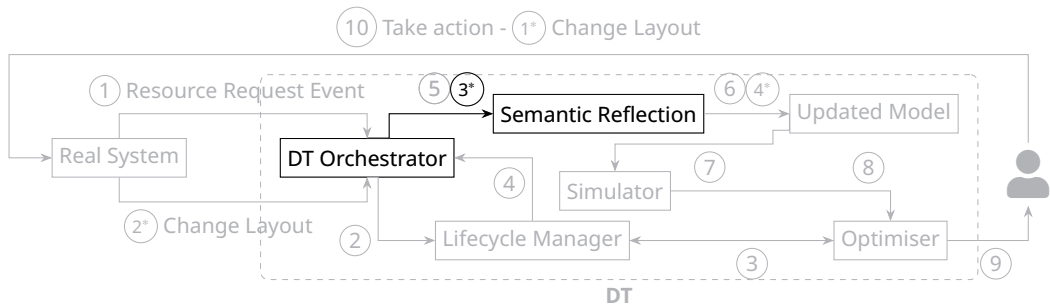
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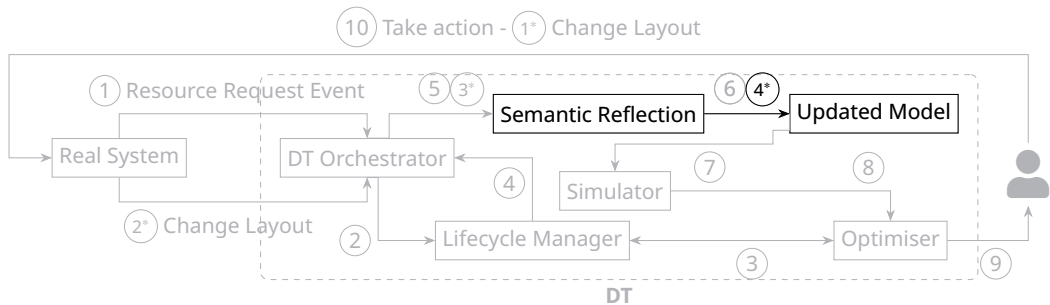
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Human-in-the-loop (cont.)

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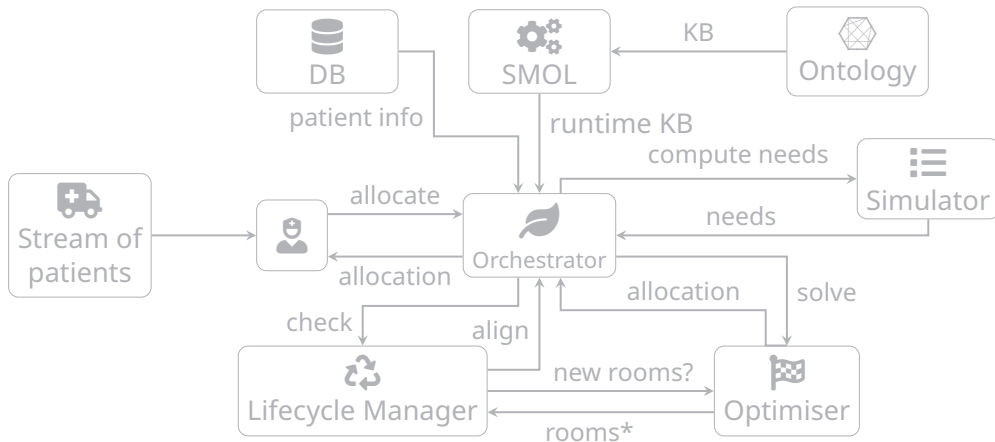
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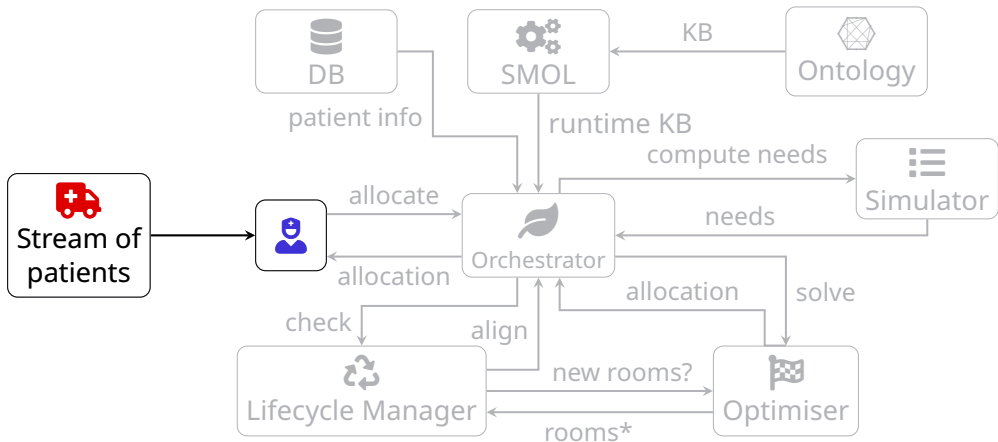
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- This enables supervised autonomy: controlled self-adaptation instead of static decision support

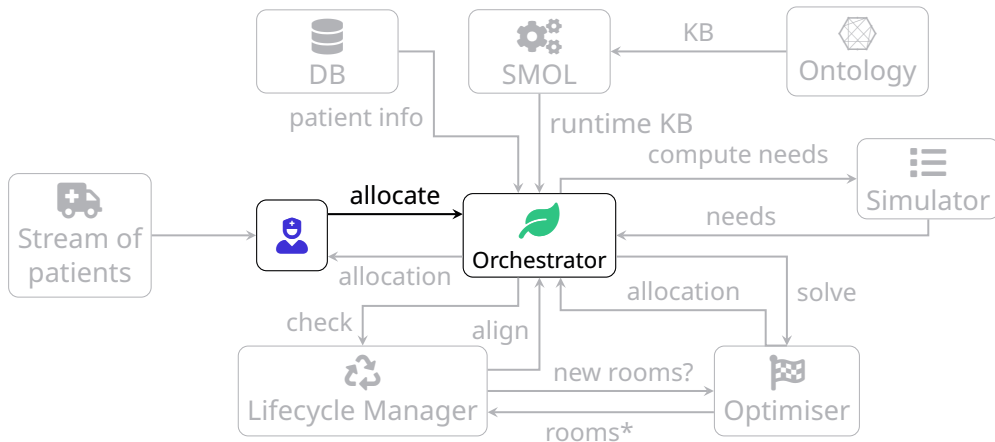
Allocation of patients



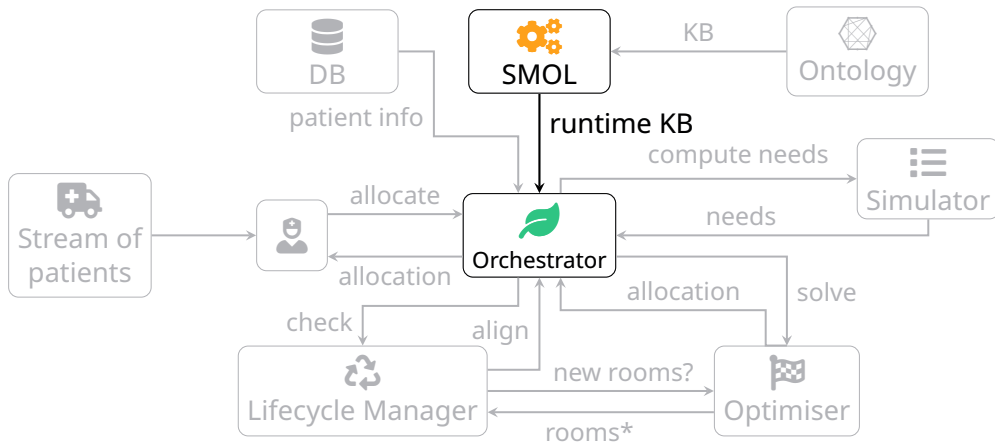
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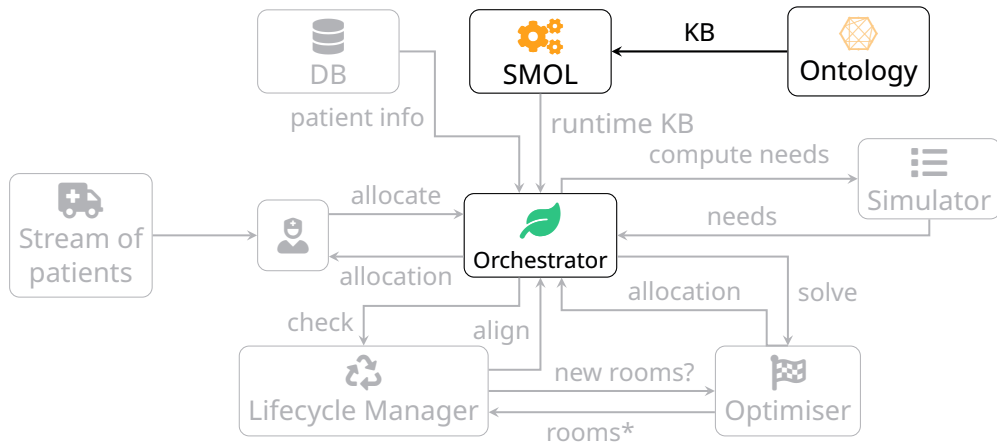
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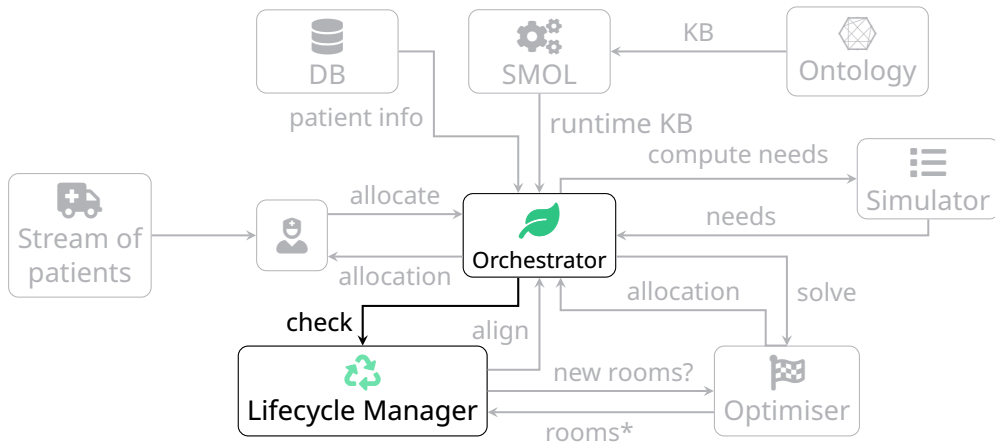
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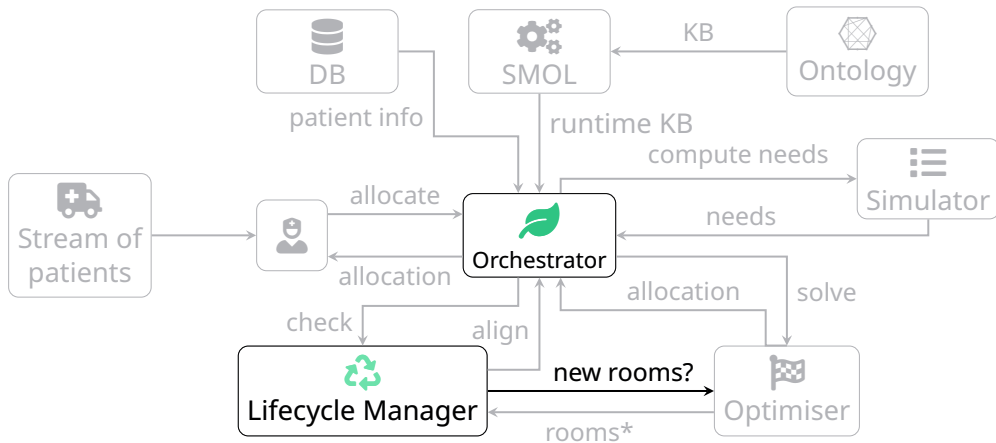
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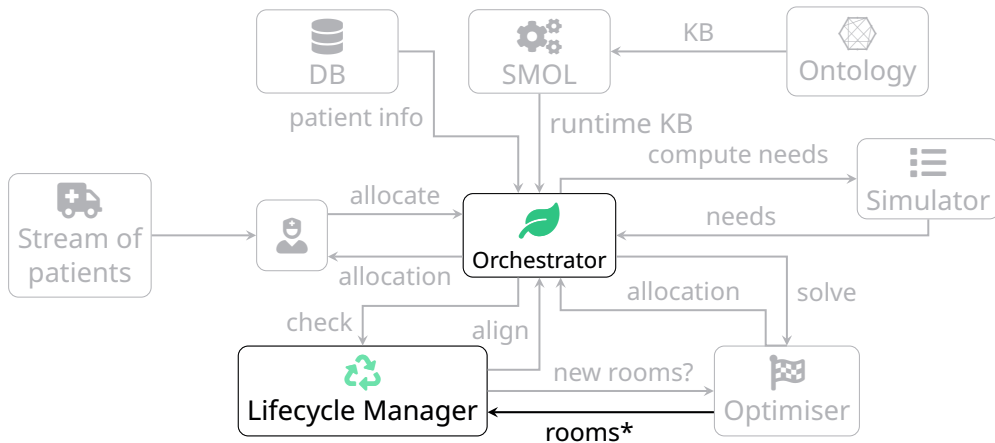
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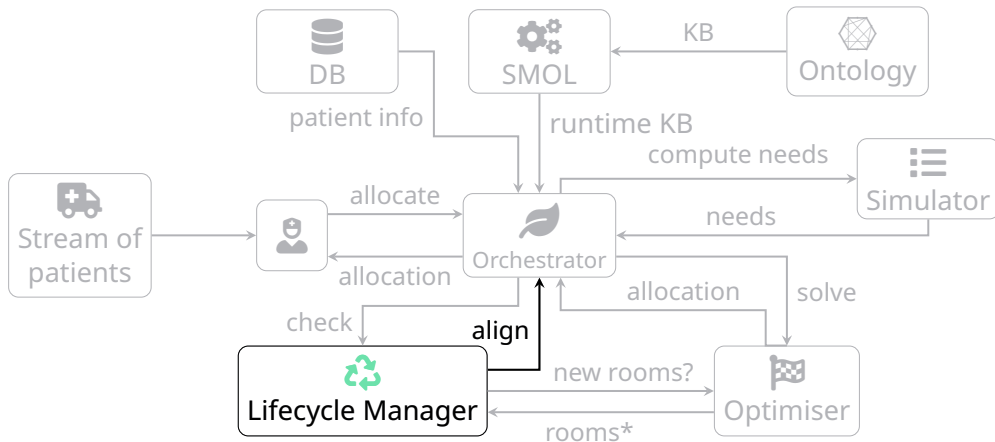
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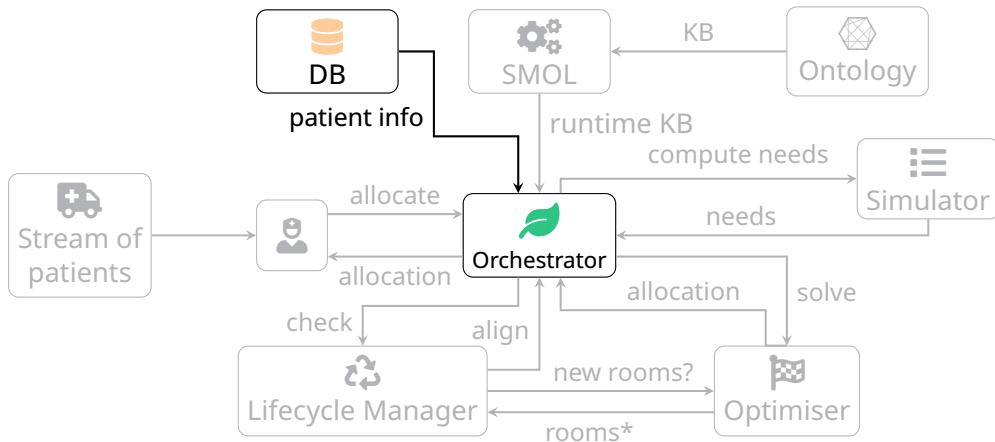
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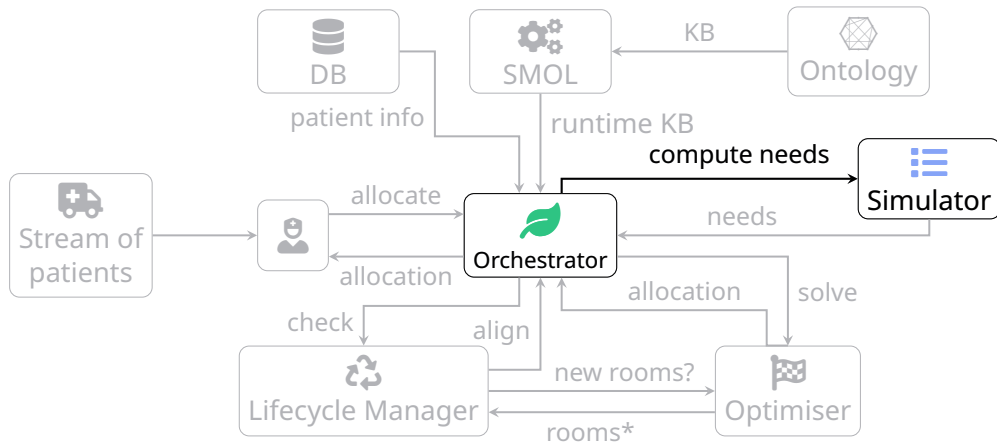
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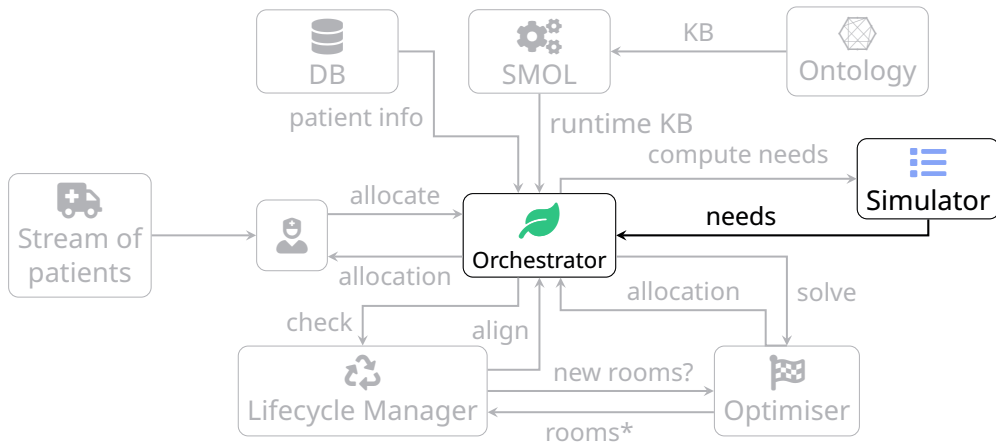
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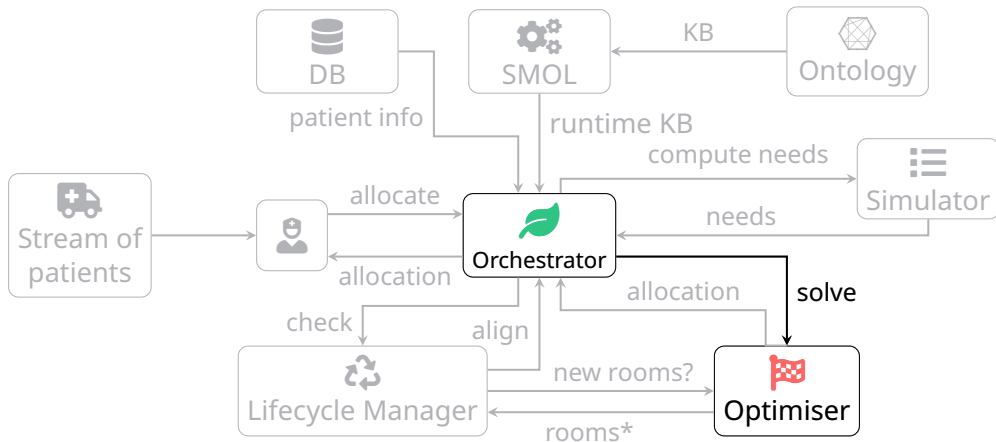
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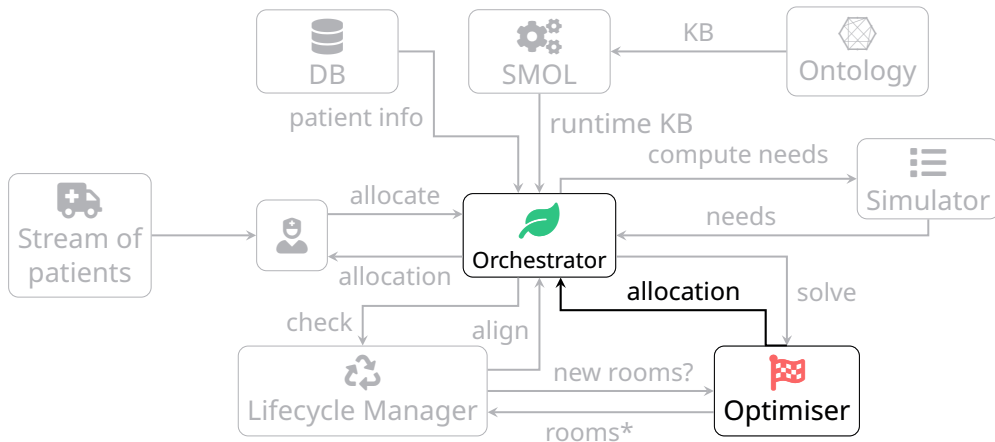
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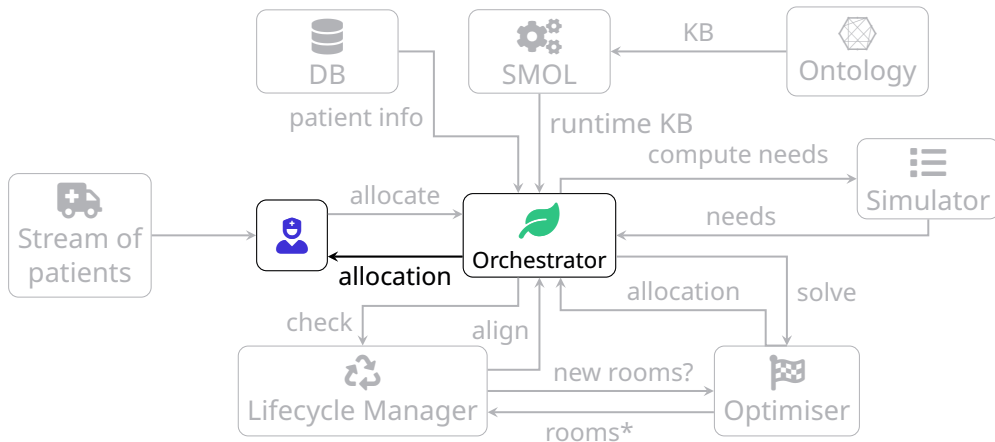
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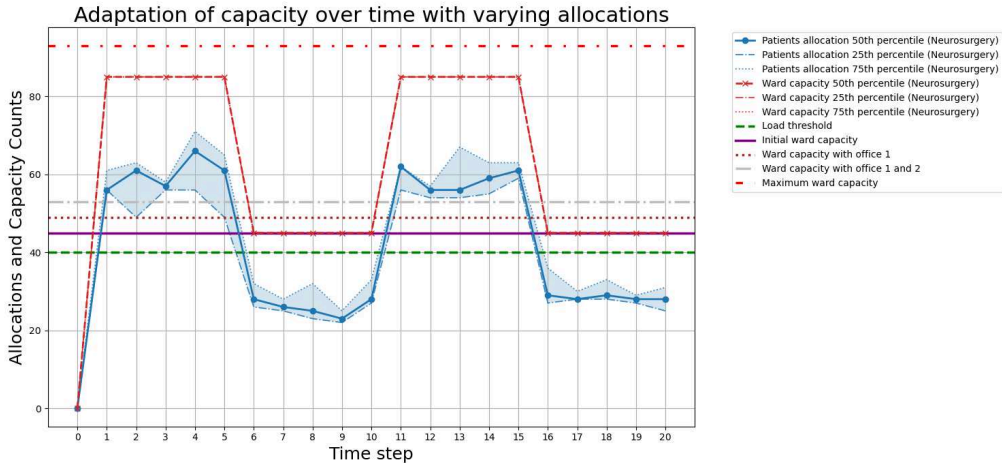
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Allocation of patients



Allocation of patients (cont.)



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- Runtime adaptation: opens/closes spare rooms in response to demand.
- Alignment: semantic lifting keeps the DT consistent after human actions.
- Cost awareness: penalties encode policy (offices before corridor)
- Trade-off: correctness/alignment vs runtime overhead
- Current limitation: thresholds and penalties are manually tuned

Future directions

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- **Learned penalties/thresholds** (data-driven tuning)
- **Live data integration** (connect to ward systems rather than simulator surrogate)
- **Broader scope** (multi-ward, staff constraints, other domains)

Thanks for the attention

Questions?

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